Group 5 : Leyla, Francesco, Daniel, Zain

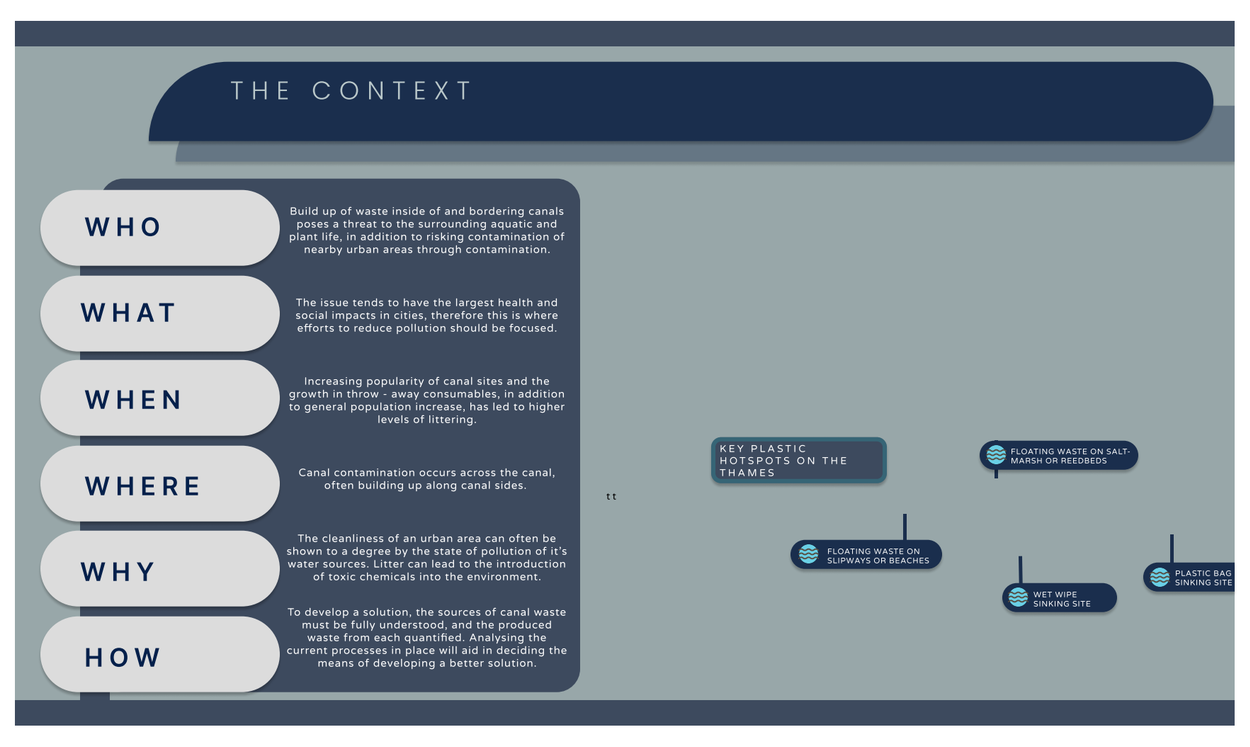
Sentire

Report.Clean.Repeat.

+ 5
POINTS
Keeping London's waters clean.
Get started

Canal & River Trust

Group 5 : Leyla, Daniel,Francesco, Zain



THE CONTEXT

WHO Build up of waste inside of and bordering canals poses a threat to the surrounding aquatic and plant life, in addition to risking contamination of nearby urban areas through contamination.

WHAT The issue tends to have the largest health and social impacts in cities, therefore this is where efforts to reduce pollution should be focused.

WHEN Increasing popularity of canal sites and the growth in throw - away consumables, in addition to general population increase, has led to higher levels of littering.

WHERE Canal contamination occurs across the canal, often building up along canal sides.

WHY The cleanliness of an urban area can often be shown to a degree by the state of pollution of it's water sources. Litter can lead to the introduction of toxic chemicals into the environment.

HOW To develop a solution, the sources of canal waste must be fully understood, and the produced waste from each quantified. Analysing the current processes in place will aid in deciding the means of developing a better solution.

KEY PLASTIC HOTSPOTS ON THE THAMES

FLOATING WASTE ON SALT-MARSH OR REEDBEDS
FLOATING WASTE ON SLIPWAYS OR BEACHES
WET WIPE SINKING SITE
PLASTIC BAG SINKING SITE

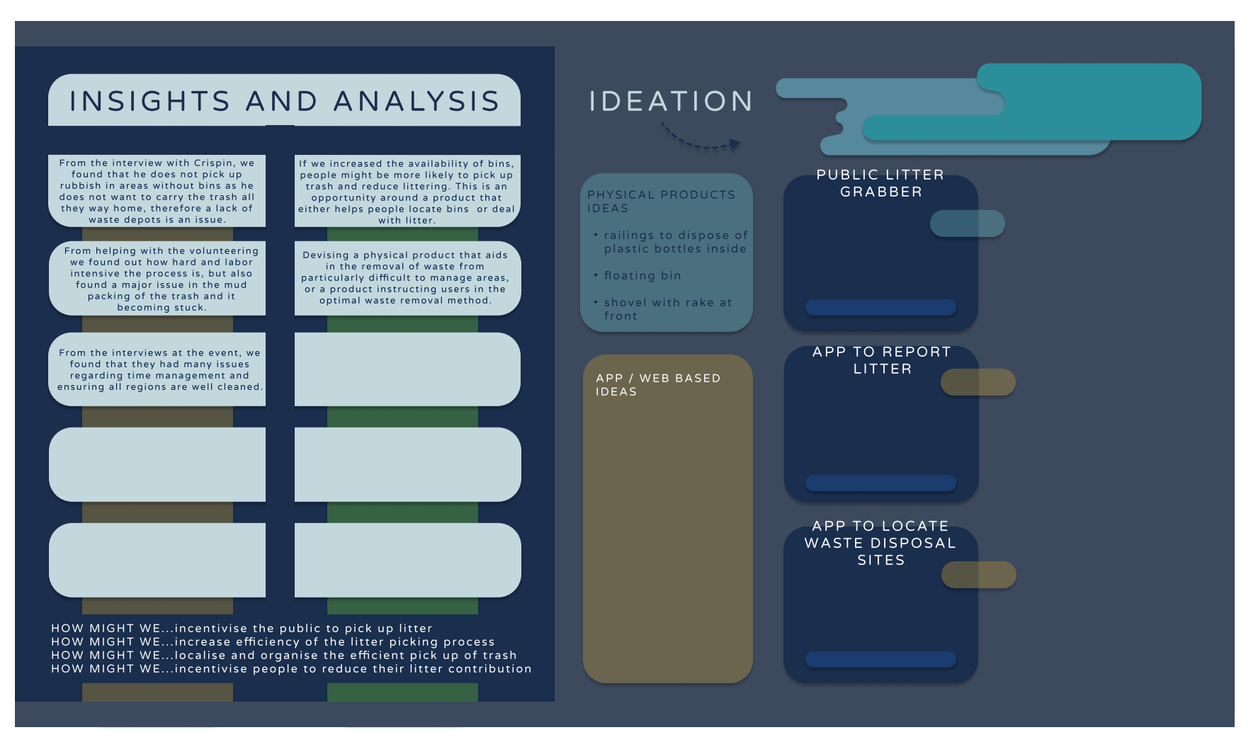
OBSERVATIONS

USER RESEARCH

WORKING WITH THE CANAL AND RIVER TRUST

THEIR CURRENT PROCESS

① ② ③ ④



INSIGHTS AND ANALYSIS

From the interview with Crispin, we found that he does not pick up rubbish in areas without bins as he does not want to carry the trash all they way home, therefore a lack of waste depots is an issue.

If we increased the availability of bins, people might be more likely to pick up trash and reduce littering. This is an opportunity around a product that either helps people locate bins or deal with litter.

From helping with the volunteering we found out how hard and labor intensive the process is, but also found a major issue in the mud packing of the trash and it becoming stuck.

Devising a physical product that aids in the removal of waste from particularly difficult to manage areas, or a product instructing users in the optimal waste removal method.

From the interviews at the event, we found that they had many issues regarding time management and ensuring all regions are well cleaned.

HOW MIGHT WE...incentivise the public to pick up litter
HOW MIGHT WE...increase efficiency of the litter picking process
HOW MIGHT WE...localise and organise the efficient pick up of trash
HOW MIGHT WE...incentivise people to reduce their litter contribution

IDEATION

PHYSICAL PRODUCTS IDEAS

- railings to dispose of plastic bottles inside
- floating bin
- shovel with rake at front

APP / WEB BASED IDEAS

PUBLIC LITTER GRABBER

APP TO REPORT LITTER

APP TO LOCATE WASTE DISPOSAL SITES

FURTHER IDEATION AND CODESIGN

USER FEEDBACK

CO-DESIGN SESSION

CANAL AND RIVER TRUST -
HOW CAN THEY BE AIDED?

1. location of the litter
2. severity of the waste
3. are there any toxic materials such as batteries or chemical bottles that can be seen?
4. does the vegetation need to be cut back?
5. is heavy machinery required?

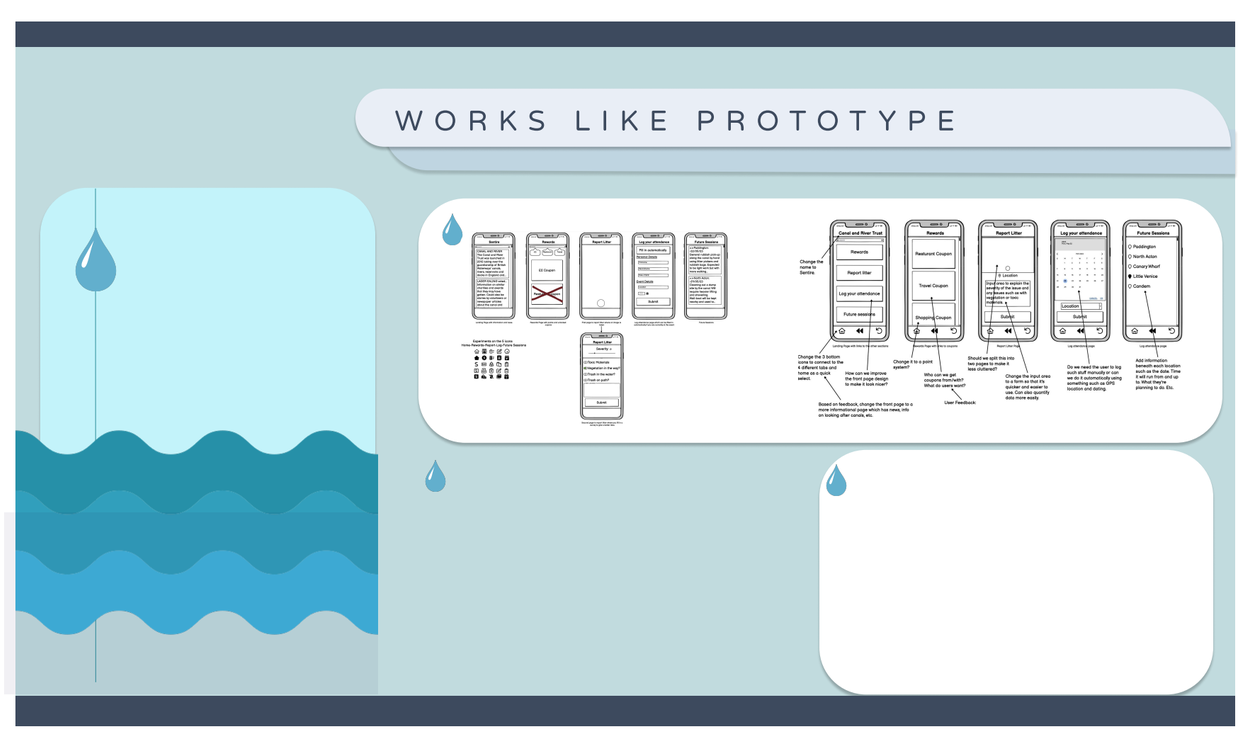
BRAND ICEBERG

VISUAL IDENTITY
ESSENCE
DIFFERENTIATION
ORIGIN

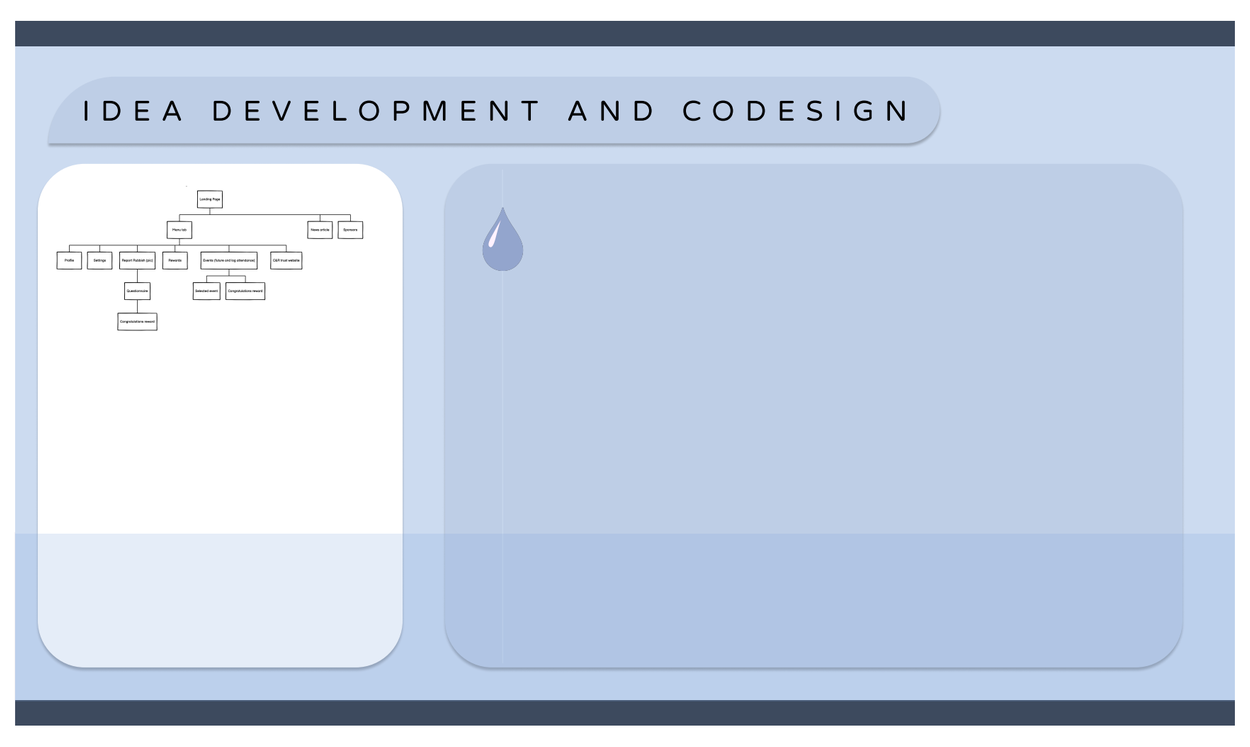
DEVELOPING OUR BRAND IDENTITY

RELATIONSHIP WITH THE CANAL AND RIVER TRUST

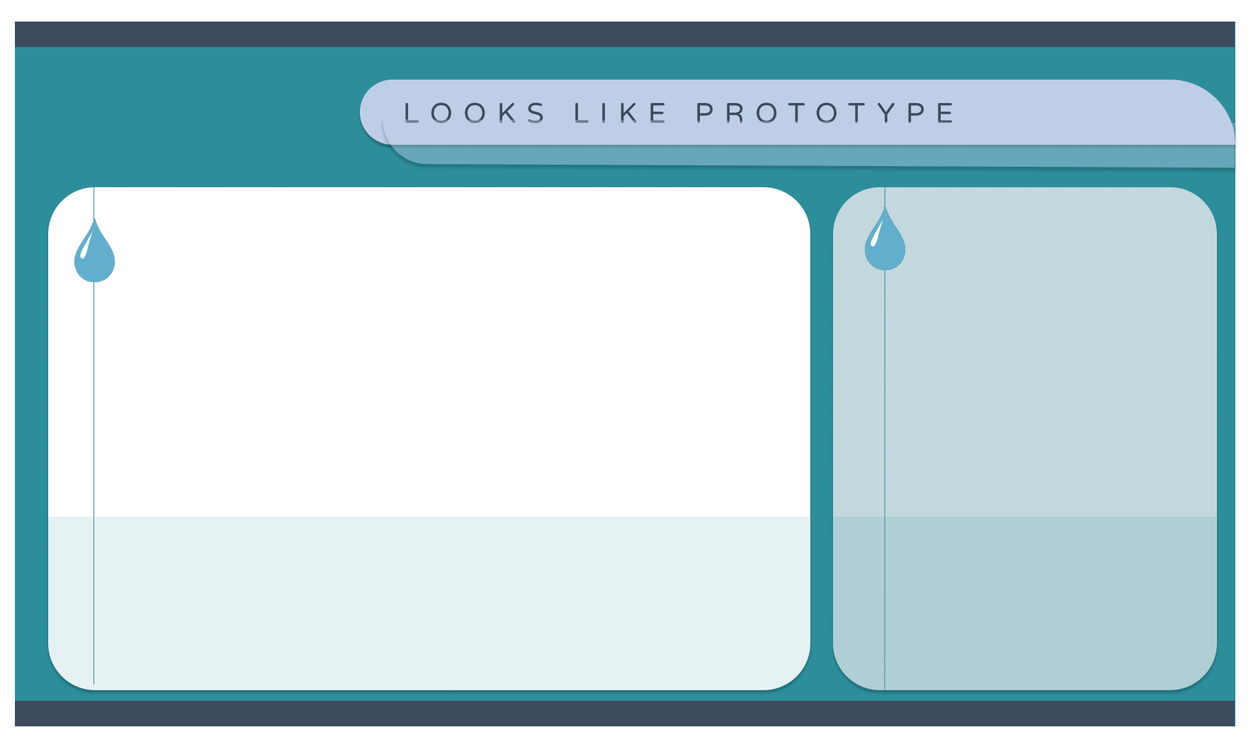
WORKING WITH THE TRUST



WORKS LIKE PROTOTYPE



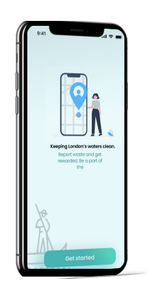
IDEA DEVELOPMENT AND CODESIGN



LOOKS LIKE PROTOTYPE

REQUIREMENTS SPECIFICATIONS VALIDATED?

FINAL PRODUCT



PROJECT DIRECTION

PROJECT BRIEF

What & Who?

Worker's of the canal and river trust find it difficult to collect information about wastage of canals, and need a more efficient method to clear up London's canals.

When & Where?

London has a network of canals thanks to the famous River Thames. The waterways of London are used on a daily basis by thousands of people, which requires constant cleaning.

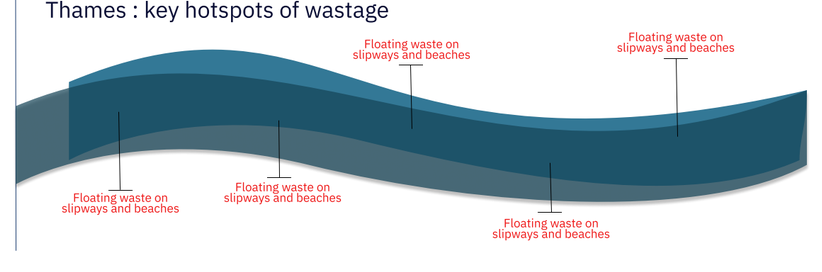
Why ?

Not only does litter render areas unsightly and dirty, it is also a major environmental hazard both through the pollution it causes and the harmful chemicals that can be found in many everyday products, such as batteries and cigarettes.

How ?

Begin with desk research and then move onto User interviews and research in order to gain a better understanding of problems the water cleaning companies face and ways they might be tackled.

Thames : key hotspots of wastage


Floating waste on slipways and beaches
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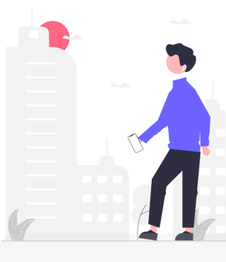


32%

London has one of the lowest recycling rates in the country -only 32% of all waste being recycled or composted and this makes waste one of the biggest environ- mental issues in London. [1]

62

62 serious pollution incidents were reported by UK water and sewage [2]

300k

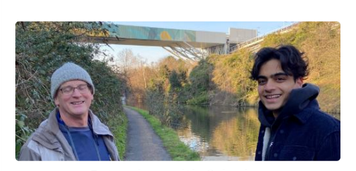
Last year, Ealing council spent over £300,000 cleaning up litter [3]

9 mn

Over 9 million people live within 9km of a canal in the UK.

USER RESEARCH

WORKING WITH THE CANAL & RIVER TRUST



Interview with Crispin

Why do people litter ?

- "The area is hidden from direct view (such as behind bushes, underneath stairs and in alleys) "
- "A lack of bins often results in people not wanting to carry trash (Crispin only picks up litters when he goes on walks if he knows there's a bin he can drop the trash off at on the way)
- "If people see litter, they are more likely to litter themselves Fly litterers do so to avoid paying money to get rid of their waste or to avoid travelling far"

Observations around the canal

Going around the canals, we found that most litter formed in areas around bushes which covered the view from direct sight, areas near exits to the canal, and areas away from bins. This allowed us to make the assumption that: **Litter seems to form in secluded and covered areas.**

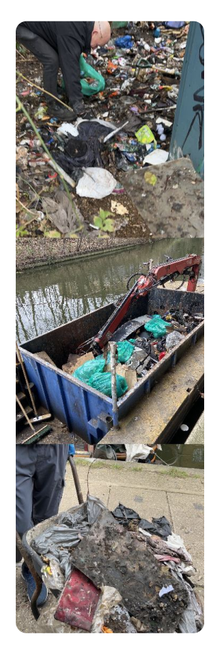
In terms of waste that we found, the vast majority of it was litter such as **cans, bottles and packaging**. However, there were also a few **cigarettes** around and every once in a while we would find something larger like a pile of clothes or a lime bike which would definitely be harder to clean up.

Types of waste

dumped litter - clothes, mattresses, kettles

stolen property - lime bikes, wallets, phones

casual litter - cans ,paper, bottles



Paddington event

Interview with **Mark :**

At this event, we met Mark who also works for LAGER Can who happily complained to us about issues he has, such as the company Kingdom Security that is supposed to stop littering in Ealing but does "f*ck-all".

Interview with **Greg**:

Greg talked to us about the process and why they do things. For example, they have found that if they make the area look clean and looked after then people are much less likely to litter. He wishes that there was a system in place like Singapore that would punish litterers more but knows that's unlikely to occur.

Their process

1. Firstly, they **identify areas with a large build-up of litter.** This is either done by public reports that are emailed to them or volunteers which spot and report the severity and position.
2. The next step is to move the welled narrow boat as close to its location as possible in order to facilitate it being filled up and to transport the equipment used for the clean-up including shovels, gloves, wheelbarrows, rakes, etc.
3. The waste is then **shovelled,** picked up, put into wheelbarrows and then unloaded into the well of the narrowboat.
4. When the narrowboat is either too full to continue, or the waste is fully cleared, it will then transport the waste to a waste depot that sorts and recycles it if able.
5. If there is a need, the vegetation would also **be trimmed down** or even fully cut down depending on the density of its foliage and how impactful it is on the walkway.

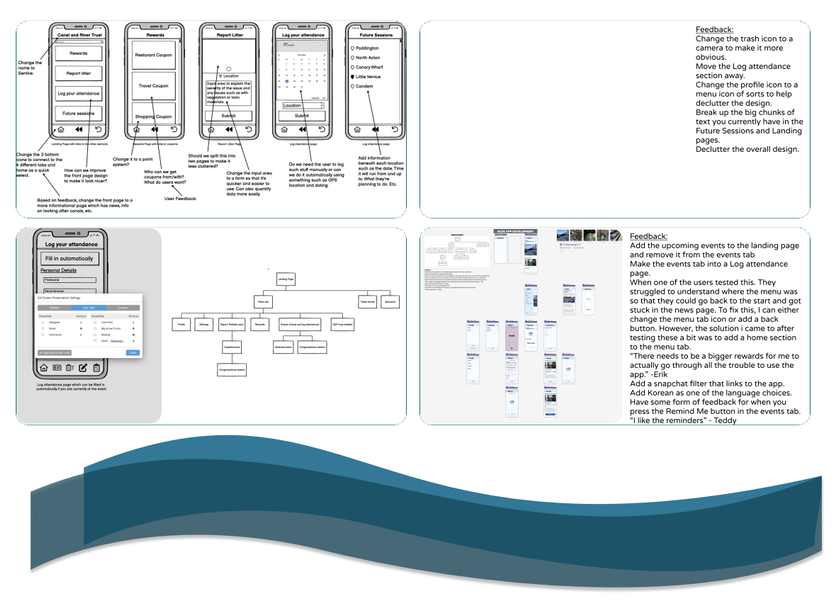
We attended an event near Paddington where we helped clean up an exceptionally bad area while conducting observations and interviewing the other volunteers.

INSIGHTS & ANALYSIS

Sentire
FOCUS CAMERA ON WASTE

WORKS LIKE PROTOTYPE

The 4 prototypes:


Log your attendance
Future Sessions
Paddington
North Acton
Canary Wharf
Little Venice
Condem
Location
Submit
Do we need the user to log such stuff manually or can we do it automatically using something such as GPS location and dating.
Add information beneath each location such as the date. Time it will run from and up to. What they're planning to do. Etc.

Feedback:
Change the trash icon to a camera to make it more obvious.
Move the Log attendance section away.
Change the profile icon to a menu icon of sorts to help declutter the design.
Break up the big chunks of text you currently have in the Future Sessions and Landing pages.
Declutter the overall design.

Feedback:
Add the upcoming events to the landing page and remove it from the events tab
Make the events tab into a Log attendance page.
When one of the users tested this. They struggled to understand where the menu was so that they could go back to the start and got stuck in the news page. To fix this, I can either change the menu tab icon or add a back button. However, the solution i came to after testing these a bit was to add a home section to the menu tab.
"There needs to be a bigger rewards for me to actually go through all the trouble to use the app." -Erik
Add a snapchat filter that links to the app.
Add Korean as one of the language choices.
Have some form of feedback for when you press the Remind Me button in the events tab.
"I like the reminders" - Teddy

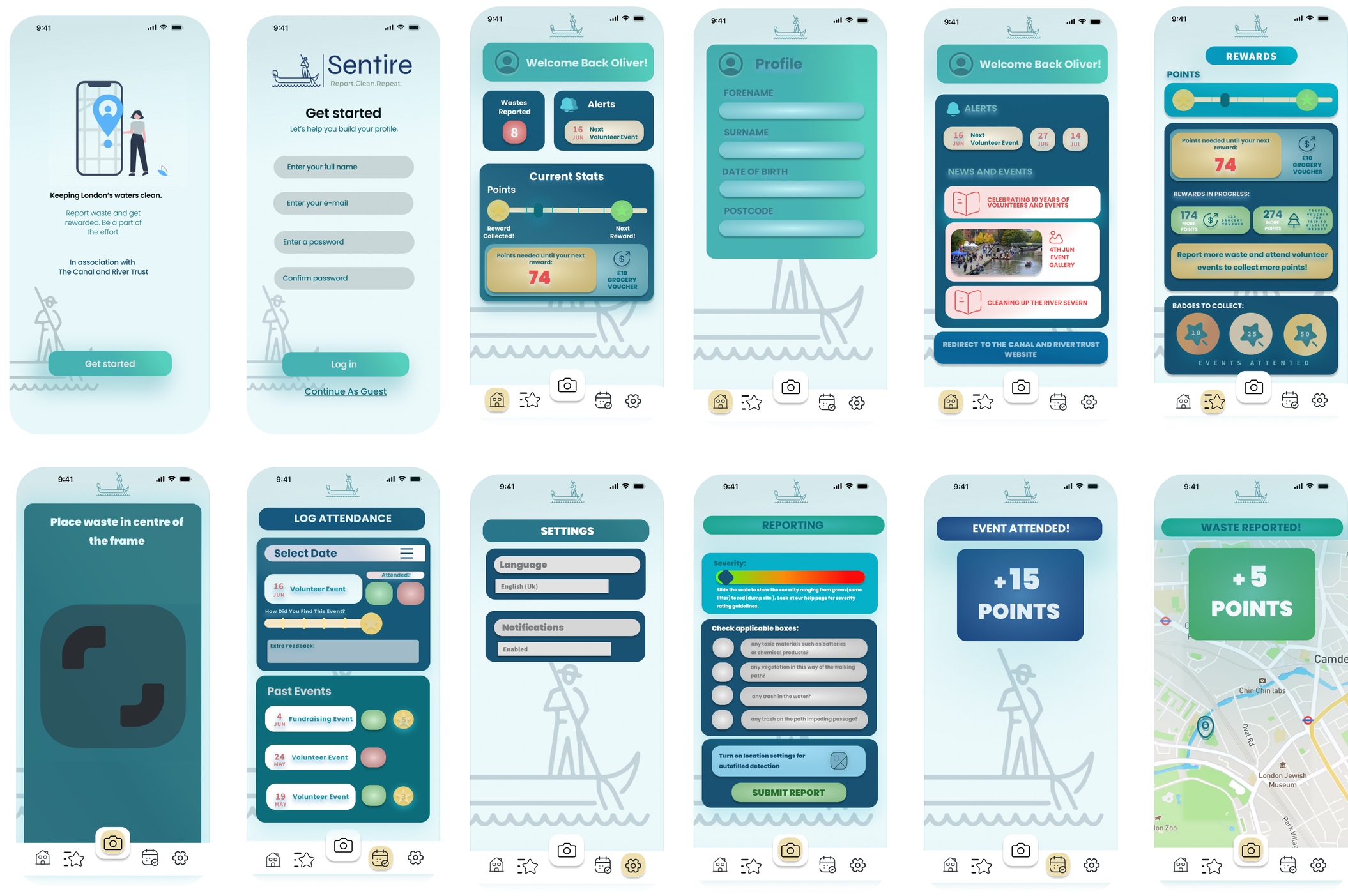
The first 3 prototypes were made using Balsamiq Wireframes.

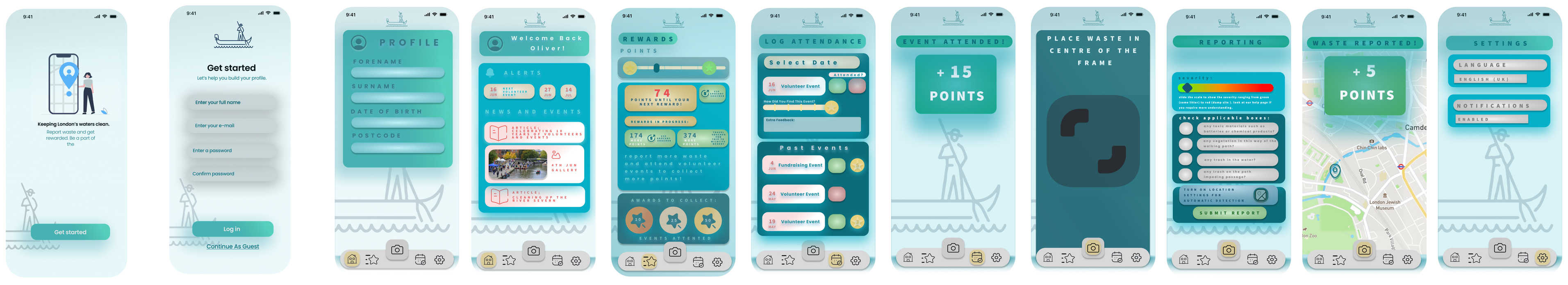
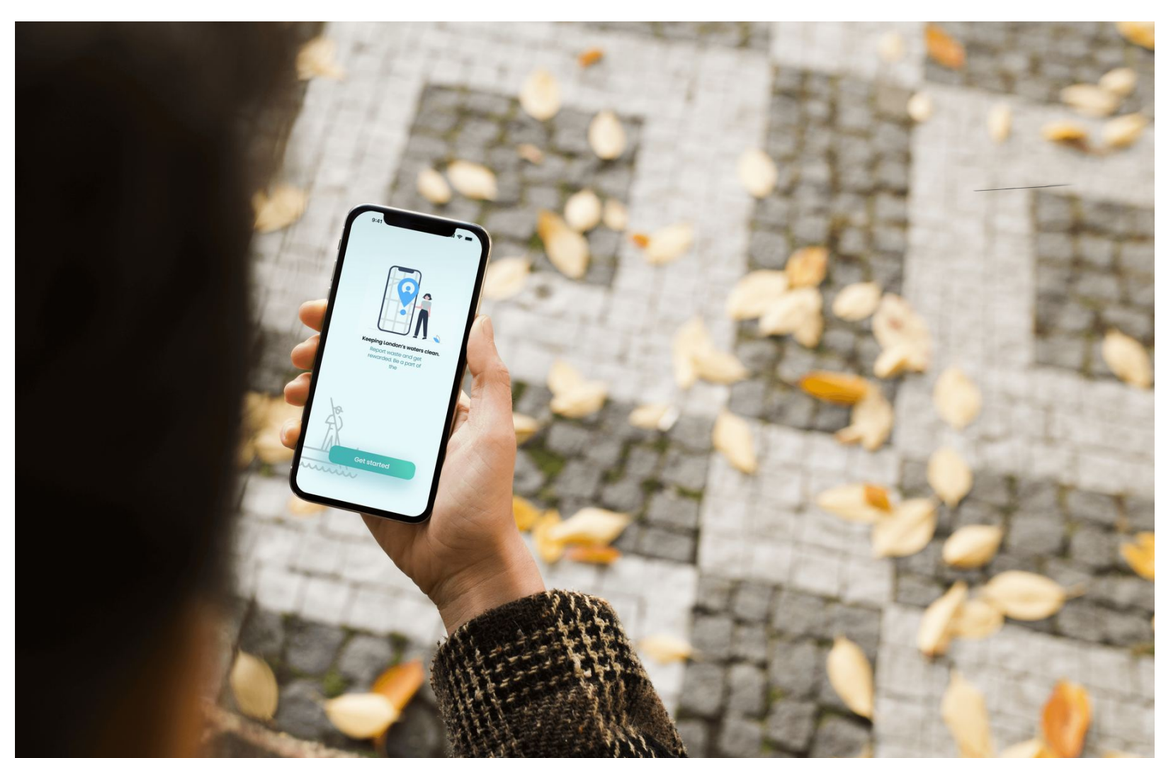
The first one was used to show the potential users the most basic layout for it and the key features that the app would need to have. The feedback from this mostly led to a change in the layout and the introduction of news to the landing page as that is something that the client asked of us.

The second iteration led to a optimisation of the design based off the feedback from the first which led to a decluttering of some pages and changes in the style. The main aim of this iteration was to see how the user interacts and which icons they found the most helpful and symbolises the different pages the most for the menu tab at the bottom.

The third version was a user testing version where we conducted some user testing and got a bit of feedback based off it. This feedback then led to the final site map.

Lastly, the fourth version was made on Figma. This was made to be interactive and allow the users to fully test the app. This version removed the menu bar ,due to the user feedback in version 3, which decluttered the overall look of the app and made it easier to use. However, a big problem we found was that the younger users would not be incentives by the coupons that the app gives. A possible solution to this would be to link the app to a digital platform and give rewards like filters or something of that nature.


Keeping London's waters clean.
Get started
Sentire
Get started
Enter your full name
Enter your e-mail
Enter a password
Confirm password
Log in
Continue As Guest
Welcome Back Oliver!
Wastes Reported
Alerts
Current Stats
Points
74
Profile
FORENAME
SURNAME
DATE OF BIRTH
POSTCODE
NEWS AND EVENTS
REDIRECT TO THE CANAL AND RIVER TRUST WEBSITE
REWARDS
BADGES TO COLLECT:
Place waste in centre of the frame
LOG ATTENDANCE
Select Date
Volunteer Event
Past Events
Fundraising Event
SETTINGS
Language
Notifications
REPORTING
Check applicable boxes:
SUBMIT REPORT
EVENT ATTENDED!
+15 POINTS
WASTE REPORTED!
+5 POINTS
Camden


Keeping London's waters clean.
Get started
Log in
Continue As Guest
PROFILE
Welcome Back Oliver!
REWARDS
LOG ATTENDANCE
EVENT ATTENDED!
+15 POINTS
PLACE WASTE IN CENTRE OF THE FRAME
REPORTING
SUBMIT REPORT
WASTE REPORTED!
+5 POINTS
SETTINGS
LANGUAGE
NOTIFICATIONS